SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**

Question Count: 5

Code of Conduct

I Y . Y o g e s h (1 9 2 5 2 5 0 0 4) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Y.Yogesh

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Total Marks:

1. Define Cloud Computing and explain four Characteristics.

Definition:

Cloud computing means using computing devices like storage, services, and software through internet instead of owning a physical computer and server.

Four Characteristics

1. **On-Demand Self-Service:**

   Users can access cloud resources whenever they need them.

2. **Broad Network Access:**

   Cloud services can be accessed through the internet using mobiles, laptops, and computers.

3. **Resource Polling:**

   Cloud providers share computing resources among multiple users.

4. **Rapid Elasticity:** Resources can be increased or decreased according to user needs.

Conclusion: Cloud computing is flexible, cost-effective, and easy to access.

2. Explain the evolution of cloud computing from hardware to internet software.

Cloud computing developed through the following stages.

1. Mainframe Computers: Large computers were used by multiple users.
2. Personal Computers: Computers become available for individual users.
3. Client - Server Computing: Clients requested services from centralized servers.
4. Distributed Computing: Multiple computers worked together to complete tasks.
5. Virtualization: One physical server was divided into multiple VM's.
6. Internet Evolution: The internet allowed users to access applications and data from anywhere.

Conclusion: Improvements in hardware, networking, virtualization and internet technology led to modern cloud computing.

3. Explain server virtualization and differentiate it from cloud computing.

Server Virtualization: Virtualization is a technology that allows one physical server to run multiple VM using software called a hypervisor.

Advantages:

1. Reduces hardware costs.
2. Improves hardware utilization.
3. Allows multiple operating systems to run on one server.
4. Makes resource management easier.

Virtualization	Cloud Computing
Creates virtual machines	Provides computing services over the internet.
Uses a hypervisor	Uses virtualization and other technologies.
Improves hardware Utilization Ex: VM Ware	Provides On-Demand resources. Ex: AWS.

4. Discuss the role of Data Centers in delivering cloud services.

A datacenter is a facility containing servers, storage devices, and networking equipment used to store and process data.

Role of Data Centers:

Data Storage: store files, databases and applications.

Data Processing: Processes information and runs applications.

Networking : Connects servers to the internet.

Security : Protects data using security systems and access controls.

Reliability : Uses backup systems to reduce service interruptio

Scalability : Allows cloud computing to add more resources when needed.

5. Differentiate IaaS, PaaS, SaaS and XaaS with examples.

1. IaaS : Infrastructure as a service.
   Provides virtual machines, storage, and networking over the internet.
   Ex: Amazon EC2.

2. PaaS: Platform as a service.
   Provides a platform for developing and deploying applications without managing physical servers.
   Ex: Google App Engine.

3. SaaS : Software as a Service
   Provides ready-to-use software through internet.
   Ex: Google Docs

4. XaaS : Everything as a Service.
   Provides different types of IT services over the internet.
   Ex : Dropbox (storage as a service).

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